Errata for *A 2 million-person, campaign-wide field experiment shows how digital advertising affects voter turnout*

The maintained replication code behind these errata is at https://github.com/active-maintenance-aec/aggarwal_etal_2023 and is archived at Zenodo at https://doi.org/10.5281/zenodo.24168257, a DOI that always resolves to the latest version.

Seven corrections to *Nature Human Behaviour* article 10.1038/s41562-022-01487-4. None of them changes a conclusion of the article. Every corrected value below is computed at render time from a full reproduction of the published analysis, held in the repository `active-maintenance-aec/aggarwal_etal_2023`. Bold marks the tokens that changed; everything else in each corrected sentence is the published wording.

1. The number of pre-treatment covariates tested for balance

> As shown in Fig. 5, the random assignment procedure generated treatment and control groups that were balanced on pre-treatment characteristics in this subset. Out of **21** covariates

Corrected: As shown in Fig. 5, the random assignment procedure generated treatment and control groups that were balanced on pre-treatment characteristics in this subset. Out of **22** covariates

Figure 5 plots twenty-two covariates and Supplementary Table 3 prints twenty-two rows. The slip is in the count the sentence states and not in the analysis: the Benjamini-Hochberg values the paragraph goes on to print, 0.262 and 0.200, are what twenty-two tests give, where the same p-values over twenty-one would give 0.250 and 0.191. The count of two significant imbalances is right, two of twenty-two at the 0.05 level is about what chance gives (1.1 expected), and everything the paragraph goes on to say about them stands.

2. The size of the treatment audience

> The full cost of the advertising campaign was US$8.9 million, spread out over a treatment audience of **1,993,216 million** (3,322,027 programme-eligible voters × our 60% match rate), amounting to US$4.46 of advertising expenditure per voter.

Corrected: The full cost of the advertising campaign was US$8.9 million, spread out over a treatment audience of **1,993,216** (3,322,027 programme-eligible voters × our 60% match rate), amounting to US$4.46 of advertising expenditure per voter.

The parenthetical states the arithmetic: 3,322,027 times a 60 per cent match rate is 1,993,216 people. The word "million" makes the figure a million times too large, and with it the sentence's own US$4.46 per voter could not be right. With the corrected audience it is: US$4.46 follows from any total between US$8,879,777 and US$8,899,709, all of which round to US$8.9 million.

3. The degrees of freedom of the balance F test

> An F-test comparing a regression of treatment status on covariates with strata fixed effects versus a restricted model predicting treatment status from strata fixed effects was non-significant ($F_{\mathbf{1999264,-18}} = 1.0266$; $P = 0.425$).

Corrected: An F-test comparing a regression of treatment status on covariates with strata fixed effects versus a restricted model predicting treatment status from strata fixed effects was non-significant ($F_{\mathbf{18,1999246}} = 1.0266$; $P = 0.425$).

An F statistic carries its numerator degrees of freedom first. The unrestricted model has 18 more parameters than the restricted one and leaves 1,999,246 residual degrees of freedom. The published subscripts are the second row of the Wald test's printed output, where the first figure is that model's residual degrees of freedom and the second is the difference in parameters, printed as a negative number. The statistic and the p-value are right. The 18 is not the 22 of entry 1 less anything a correction touches. Party and Trump support score band are each exhaustive sets of indicators, so one of each is the reference category, and the authors' model also leaves out the ideology and partisanship scores. With those two included the F statistic is 0.9763 on 20 and 1,999,244 degrees of freedom ($P = 0.488$), so "we could not predict treatment assignment from the covariates" holds over every covariate Figure 5 reports.

4. Acronym's share of federal advertising spending on Facebook

> According to this data, there was a total of approximately $977,761,865 spent on federal races on Facebook from 2/1/2020 to 11/3/2020, $6,603,488 (**0.67%**) of which was from Acronym.

Corrected: According to this data, there was a total of approximately $977,761,865 spent on federal races on Facebook from 2/1/2020 to 11/3/2020, $6,603,488 (**0.68%**) of which was from Acronym.

A rounding artifact rather than a substantive error: the quotient of the sentence's own two figures is 0.6754 per cent, which rounds up. The share is kept at two decimal places because at one decimal the published and corrected figures are the same. What the share supports is the next paragraph's "relatively low percentage of total spending", which either figure does.

5. The definition of the black variable in the randomisation code

> , case when vb_voterbase_race = 'African-American' or (vb_voterbase_race = 'Uncoded' and **civis_race**

Corrected: , case when vb_voterbase_race = 'African-American' or (vb_voterbase_race = 'Uncoded' and **civis_race = 'AFAM') then 'black' else null end as black**

The line is drawn to x 610.6 on a page 612 points wide and the text layer ends with `civis_race`. The complete line is in the authors' own `Acronym/notes/gh_randomization_sql.mssql`, line 17, which agrees with every other line of the published listing except for the capitalisation of the schema name. `black` is one of the two race categories the Methods report as no longer available, and the code as printed cannot be run. No quantity in the article rests on the line, but it defines a stratum. Step 4 of the assignment procedure in Fig. 4 samples 327,230 "of (3,252,002 + ?) Black voters", the listing draws them `where black = 'black'`, and the clause the published page cut off is what the `+ ?` counts: uncoded voters whom a third-party imputation classified as Black. That imputation is the lost data that leaves the denominators of steps 4 and 5 uncomputable, and the restored line shows exactly where it entered.

6. The definition of the hispanic variable in the randomisation code

> , case when vb_voterbase_race = 'Hispanic' or (vb_voterbase_race = 'Uncoded' and **civis_race = 'HISPA**

Corrected: , case when vb_voterbase_race = 'Hispanic' or (vb_voterbase_race = 'Uncoded' and **civis_race = 'HISPANIC') then 'hispanic' else null end as hispanic**

The line is drawn to x 615.8 on a page 612 points wide, so it ends beyond the page edge and mid-token, at `'HISPA`. The complete line is in the authors' own `Acronym/notes/gh_randomization_sql.mssql`, line 18. `hispanic` is the other race category the Methods report as no longer available. Like entry 5, it defines a stratum rather than a quantity: step 5 of Fig. 4 samples 68,842 "of (1,267,156 + ?) Latinx voters", and the restored clause is what the `+ ?` counts.

7. The month of heaviest persuasion spending

> In fact, we observe significant early spending (Acronym spent more on persuasion in **July** than in any other month, see main Figure **??**), and we find much stronger effects on early voting than day of voting (see Figure 3 in the main text).

Corrected: In fact, we observe significant early spending (Acronym spent more on persuasion in **February** than in any other month, see **Supplementary Figure 2**), and we find much stronger effects on early voting than day of voting (see Figure 3 in the main text).

The figure the sentence points at is Supplementary Figure 2, whose two panels split Acronym's monthly lower-bound spending on persuasion advertisements by format and by keyword. Summed over either panel's series, the largest month is February 2020 at US$577,000, against US$566,600 in July. The cross-reference did not resolve when the document was typeset. The Ad Library reports each advertisement's spend as a range, dated to the month its delivery began, and the ranking holds only at the lower end the figure plots: at the upper end February is fourth, at US$721,730, behind

October (US$963,603), September (US$900,695), and July (US$745,783), and it is fourth on the midpoints too. The corrected sentence is therefore true of the figure it cites, and the data support no more than that Acronym spent heavily from February onward. That is what the sentence's argument needs: its claim is that Acronym spent earlier in the campaign than other left-leaning advertisers did, and February is earlier than July.